domain of 100 electrons aligned spin-up next to another domain of 100 electrons all aligned spin-down. So at the very local level, electrons minimize their energy by being aligned, but when too many are aligned and the magnetic field becomes too strong, the overall energy of the system grows, causing the electrons to align only into relatively small domains.

Presumably the interactions between trillions of electrons in the bulk material result in the complex organization of electrons into domains, but it is very difficult to model that many interactions. So physicists made a simplifying assumption that a given electron is only affected by its nearest neighbors, which is exactly the same assumption we've made with neighborhood Q-learning (figure 9.10).

Figure 9.10. This is a high-resolution Ising model where each pixel represents an electron. The lighter pixels are spin-up, and black is spin-down. You can see that the electrons organize into domains where all the electrons within a domain are aligned, but nearby electrons in an adjacent domain are anti-aligned with respect to the first domain. This organization reduces the energy of the system.

